Description

Image







Caption

1. Samples of silicon carbide sandpaper. © Tiesse at en.wikipedia - Public domain 2. Silicon carbide grinding wheel. 3. U.S. Navy technician uses a silicon carbide sander. © U.S. Navy - Public domain

The material

Silicon carbide (SiC, carborundum), made by fusing sand and coke at 2200 C, is the grit on high quality sandpaper. It is very hard and maintains its strength to 1400C high temperature, has good thermal shock resistance, excellent abrasion resistance, but, like all ceramics, it is brittle. It has the highest corrosion resistance of all advanced ceramics.

Compositional summary

SiC

General properties

Density		3e3	-	3.2e3	kg/m^3
Price	*	10.9	-	15.6	GBP/kg
Date first used		1893			

Mechanical properties

Young's modulus		390	-	410	GPa
Shear modulus	*	171	-	180	GPa
Bulk modulus	*	181	-	190	GPa
Poisson's ratio	*	0.13	-	0.15	
Yield strength (elastic limit)	*	305	-	336	MPa
Tensile strength	*	305	-	336	MPa
Compressive strength	*	3.05e3	-	3.36e3	MPa
Elongation	*	0.07	-	0.09	% strain

Hardness - Vickers		2.38e3	-	2.63e3	HV
Fatigue strength at 10^7 cycles	*	259	-	302	MPa
Fracture toughness		2.1	-	2.5	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	2e-5	-	5e-5	

Thermal properties

Melting point		2.15e3	-	2.25e3	°C
Maximum service temperature		1.47e3	-	1.54e3	°C
Minimum service temperature		-273			°C
Thermal conductor or insulator?		Good conductor			
Thermal conductivity		100	-	150	W/m.°C
Specific heat capacity		663	-	677	J/kg.°C
Thermal expansion coefficient		2.7	-	2.8	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Poor insulator			
Electrical resistivity	*	1e9	-	3.16e10	µohm.cm
Dielectric constant (relative permittivity)	*	7	-	9	
Dissipation factor (dielectric loss tangent)	*	0.001	-	0.005	
Dielectric strength (dielectric breakdown)	*	5	-	10	MV/m

Optical properties

Transparency		Opaque			
Refractive index		2.66	-	2.7	

Critical Materials Risk

High critical material risk?		No			

Processability

Moldability		2	-	3	
Machinability		1	-	2	

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Excellent			
Soils, alkaline (clay)		Excellent			
Wine		Excellent			

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Excellent
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Excellent
Hydrofluoric acid (40%)	Excellent
Nitric acid (10%)	Excellent
Nitric acid (70%)	Excellent
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Excellent
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Excellent

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Excellent
Acetone	Excellent
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent

Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Acceptable
O2 (oxygen gas)	Excellent
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Excellent
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Excellent
Tolerance up to 450 C (842 F)	Excellent
Tolerance up to 850 C (1562 F)	Excellent
Tolerance above 850 C (1562 F)	Excellent

Geo-economic data for principal component

Annual world production, principal component	1.01e6	-	2.75e6	tonne/yr

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 6.92	-	7.63	kg/kg
Embodied energy (virgin grade production)	* 115	-	127	MJ/kg
Water usage	* 33.5	-	101	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 311	-	354	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	* 20.4	-	23.2	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 5.54	-	6.1	kg/kg
Embodied energy, recycling	* 92	-	102	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Technical ceramics that are used in the pure state, as SiC usually is, are very energy intensive. The ingredients, silicon and carbon, are plentiful, but processing costs make the product expensive.

Supporting information

Design guidelines

Silicon carbide and silicon nitride are two of the emerging breed of high performance technical ceramics. Their extreme corrosion resistance and high hardness makes them a good choice for mechanical components that must withstand corrosive fluids - bearings, including ball bearings, and valve and pump parts in sewage systems, for example. Their other unique feature is their ability to carry significant loads at temperatures as high as 1800 C. The main drawbacks are their low toughness, requiring careful design and flaw-free fabrication, and their high cost, which has slowed their take up. Technical ceramics are formed by the following steps.(a) Pressing, isostatic pressing, powder extrusion (for bars and tubes) or powder injection molding (for intricate, high-volume parts).(b) Green-machining in the unfired state, using standard tools.(c) Firing or "sintering" typically at 1550 - 1700 C for 12 to 20 hours; the part shrinks by about 20%.(d) Diamond grinding to achieve tighter tolerance and surface finish: +/- 10 microns is achievable. The cost of a ceramic part is greatly increased if it has to be diamond-ground. Thus design for net-shape sintering, eliminating step (d) is highly desirable. The standard tolerance for as-fired dimensions is +/- 1% or 125 microns, whichever is greater. Silicon carbide is a blue-black in color; silicon nitride is dark gray or black. Both can be polished to a very smooth, reflective surface, giving parts with a striking appearance.

Technical notes

Silicon carbide starts as a powder, is pressed (with a polymer binder) to the desired shape, then fired at a high temperature, burning off the binder and causing the powder to sinter. It is exceptionally wear and corrosion resistant. Its electrical properties can be adjusted by doping. High strength SiC fibers such as Nicalon, made by CVD processes, are used as reinforcement in ceramic and metal matrix composites.

Typical uses

Mechanical seal faces, bearings, turbocharger bearings, gas turbine rotors, wear and corrosion-resistant parts, high temperature devices, laboratory test equipment, hydraulic plungers, pistons, cylinder liners, guides and feeds, heating elements, body and aircraft armor.

Links

ProcessUniverse

Producers

Reference

Values marked * are estimates.
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